Your Model (Logistic)

$P(Y = \text{orange} \mid x) = \exp(b0 + b1x) / (1 + \exp(b0 + b1x))$

Friend's Model (Two-class Softmax)

$P(Y = \text{orange} \mid x) = \exp(a_{\text{orange}0} + a_{\text{orange}1}x) / [\exp(a_{\text{orange}0} + a_{\text{orange}1}x) + \exp(a_{\text{apple}0} + a_{\text{apple}1}x)]$

(a) What is the log-odds of orange versus apple in your model?

$\log[p / (1 - p)] = b0 + b1x$

(b) What is the log-odds of orange versus apple in your friend's model?

$\log[p / (1 - p)] = (a_{\text{orange}0} - a_{\text{apple}0}) + (a_{\text{orange}1} - a_{\text{apple}1})x$

(c) Suppose that in your model, $b0 = 2$ and $b1 = -1$. What are the coefficient values in your friend's model? Be as specific as possible.

Any set satisfying
$(a_{\text{orange}0} - a_{\text{apple}0}) = 2$, $(a_{\text{orange}1} - a_{\text{apple}1}) = -1$, Example (set apple as baseline):
$a_{\text{apple}0} = 0$, $a_{\text{apple}1} = 0$
$a_{\text{orange}0} = 2$, $a_{\text{orange}1} = -1$

Only the *difference* between coefficients affects predictions; adding a constant to both class scores changes nothing.

(d) Now suppose that you and your friend fit the same two models on a different data set. This time, your friend gets the estimates: $a_{\text{orange}0} = 1.2$, $a_{\text{orange}1} = -2$, $a_{\text{apple}0} = 3$, $a_{\text{apple}1} = 0.6$. What are the coefficient values in your model?

$a_{\text{orange}0} = 1.2$, $a_{\text{orange}1} = -2$
$a_{\text{apple}0} = 3$, $a_{\text{apple}1} = 0.6$

Then: $b0 = a_{\text{orange}0} - a_{\text{apple}0} = 1.2 - 3 = -1.8$
$b1 = a_{\text{orange}1} - a_{\text{apple}1} = -2 - 0.6 = -2.6$

(e) Finally, suppose you apply both models from (d) to a data set with 2,000 test observations. What fraction of the time do you expect the predicted class labels from your model to agree with those from your friend's model? Explain your answer.

Because both models have identical probabilities, their predicted labels should be identical.